Hallucination Guardrails via MFA and EVT

MFA: Mixture of Factor Analyzers

EVT: Extreme Value Theory

GMM: Gaussian Mixture Model

Project Title: Latent Manifold Density Estimation via Mixture of Factor Analyzers for Out-of-Distribution Hallucination Guardrails in RAG

1.Problem Statement

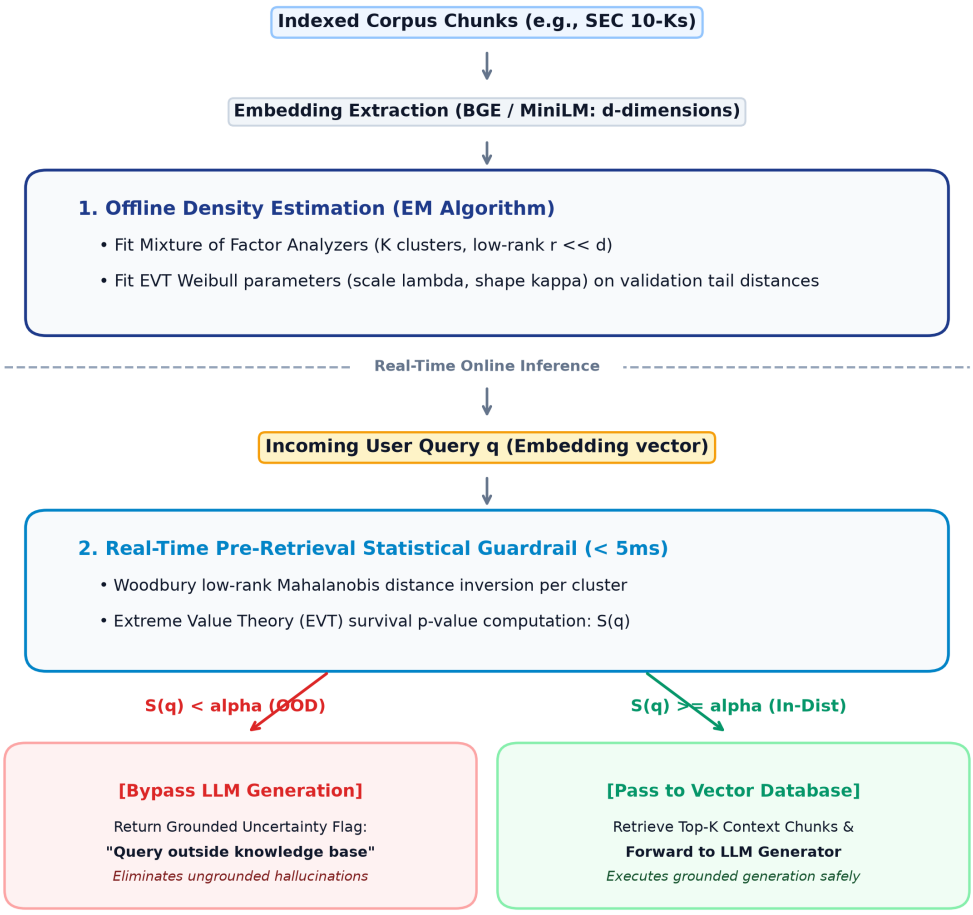
Retrieval-Augmented Generation (RAG) systems fail silently when presented with unanswerable or Out-of-Distribution (OOD) queries. Because vector databases (e.g., FAISS, ChromaDB) perform nearest-neighbor searches deterministically, they force-retrieve the closest context chunks even when a query lies entirely outside the indexed knowledge base. Conditioned on these irrelevant chunks, the Large Language Model (LLM) hallucinates plausible yet ungrounded facts.

Standard heuristics (such as raw cosine similarity cutoffs or K-Means clustering) fail in high-dimensional embedding spaces because embeddings lie along complex, anisotropic submanifolds. Furthermore, full-covariance Gaussian Mixture Models (GMMs) cannot be computed due to singularity and the curse of dimensionality (O(K*d^2) parameters).

Objective: To develop, derive, and benchmark a statistical pre-retrieval guardrail that uses a Mixture of Factor Analyzers (MFA) with Woodbury low-rank matrix inversion and Extreme Value Theory (EVT) to detect and reject OOD queries in embedding space before retrieval and LLM inference occur.

---

2. Theoretical Formulation & Architecture



---

3. Dataset & Benchmark Setup

In-Distribution (ID) Corpus Options & Sources:

- **SEC 10-K Filings (FY 2022–2023):** Sourced from the official SEC EDGAR database (~8,000–10,000 text chunks):
- SEC EDGAR Company Search
- Apple Inc. (AAPL) 10-K
- Microsoft Corp. (MSFT) 10-K
- Alphabet Inc. (GOOGL) 10-K
- **Alternative Technical Corpus:** Python 3.12 Official Documentation

Test Evaluation Sets & Sources:

Set	Count	Description / Data Source
ID Test Set	150 queries	Direct factual questions answerable strictly from the corpus.
Near-OOD / Semantic Drift	100 queries	Queries using identical vocabulary and entities but requesting unmentioned events or unreleased features.
Far-OOD	100 queries	General-knowledge QA benchmarks: HotpotQA / HuggingFace HotpotQA and TriviaQA.

Comparative Baselines:

1. Cosine similarity threshold to nearest neighbor (standard FAISS baseline)
2. Euclidean distance to nearest K-Means centroid
3. Diagonal-covariance GMM log-likelihood cutoff

---

4. Four-Week Milestone Plan

Week	Focus	Key Deliverable
Week 1	Data Pipeline & Baselines	Corpus chunked, embedded, FAISS indexed, and query splits curated
Week 2	MFA & EM Implementation	Custom NumPy MFA with Woodbury inversion fully implemented
Week 3	EVT Calibration & Integration	Weibull tail fitted, LangChain guardrail retriever operational
Week 4	Benchmarking & Report	AUROC, FPR@95, Ragas faithfulness scores, latency profiles, final report

---